Naman Gupta

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Education

University of Southern California — Los Angeles, USA
MS in Computer Science — Jan 2026 – Dec 2027

Graphic Era Deemed to be University — Dehradun, India
B.Tech in Computer Science and Engineering — Sep 2021 – June 2025

Internship/Work Experience

Morgan Stanley — Mumbai, India

Technology Analyst — Aug 2025 – Nov 2025

- Engineered a backend performance overhaul by re-architecting query paths and caching layers across critical internal applications, reducing average response time by **30%** and improving system uptime.
- Developed a suite of reusable UI components for real-time monitoring dashboards by shipping internal tooling across engineering and operations teams, eliminating **15+** hours of manual reporting work per week for stakeholders.

Technology Apprentice — July 2024 – Aug 2025

- Designed a data-driven loan routing engine by building a recommendation model that surfaces the top 3 best-fit employees per case, cutting average task completion time by **20%** across loan processing teams.
- Built predictive workload distribution models trained on **12** months of throughput and staffing data, raising balanced resource allocation accuracy across teams by **25%** while reducing employee burnout risk.

Research Work

Deep Learning Approach for Malicious URL Detection | IEEE Paper — Feb. 2025

- Engineered a malicious URL detection system by training and benchmarking four deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) on a dataset of over **450,000** labeled URLs, hitting **99.25%** precision with Bi-LSTM.
- Conducted a comparative architecture analysis by benchmarking bidirectional against unidirectional context for sequential URL pattern recognition, improving classification F1-score by **4.8%** for more robust phishing detection.

Technical Skills

Languages: C, C++, Java, JavaScript, TypeScript, Python, SQL, HTML, CSS
Developer Tools: VS Code, Git, GitHub, Docker, Postman, Linux, Microsoft, PyCharm, Jupyter Notebook
Framework / Library: Node.js, ReactJS, FastAPI, Flask, Django, TensorFlow, Keras, Pandas, OpenCV
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Natural Language Processing, Agentic AI

Projects

AI-Powered Personal Research Assistant GitHub | FastAPI, Next.js, Postgres, Gemini — June 2026 – July 2026

- Architected a full-stack RAG research platform by pairing a FastAPI/Postgres backend with pgvector and a Next.js frontend, powering summarization, semantic chat, and comparison across **10+** integrated modules.
- Benchmarked the ingestion pipeline end-to-end across real Gemini and Voyage calls, achieving under **4.3** seconds total processing per paper, with Gemini calls alone accounting for **85-95%** of that latency.

TechHire: AI Job Search Platform GitHub | Python, FastAPI, React, PostgreSQL — May 2026 – June 2026

- Built a full-stack tech job search platform by integrating a Python scraper with a FastAPI backend and React frontend, indexing over **5,000** live postings with sub-second search across skills and salary filters.
- Integrated a multi-model Groq LLM pipeline into an AI resume analyzer that synthesizes job summaries across three models, substantially improving candidate-job fit scoring and resume quality feedback.

DNA Sequence Alignment: DP vs. Divide-and-Conquer GitHub | Python, Algorithms — Apr 2026 – May 2026

- Implemented a novel DNA sequence alignment solution using the Needleman-Wunsch algorithm, building a full **O(m x n)** dynamic programming table with configurable gap penalties and a custom mismatch cost matrix.
- Optimized memory usage by engineering Hirschberg's divide-and-conquer algorithm, cutting peak memory footprint by **60%** from **O(m x n)** down to **O(m + n)** without sacrificing alignment accuracy or traceback correctness.

Real-Time Sign Language Detection GitHub | Python, OpenCV, TensorFlow, Keras — Feb 2026 – March 2026

- Developed a real-time computer vision pipeline leveraging OpenCV for gesture capture, processing live hand gesture video streams at **30** frames per second with minimal detection latency and jitter.
- Trained a TensorFlow/Keras CNN by fine-tuning across **26** gesture classes, achieving **98%** real-time translation accuracy for practical, everyday sign language communication and use.